THE RIEMANN ZETA FUNCTION IS NOT STEPANOV ALMOST PERIODIC IN ANY SUB-STRIP OF THE CRITICAL STRIP

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ABSTRACT. For $\sigma > 1$ the Riemann zeta function is Bohr almost periodic on vertical lines, and inside the critical strip it is almost periodic in the Besicovitch B^2 sense. Between the uniform and the Besicovitch scales sits the Stepanov scale, the only one of the classical almost-periodicity scales whose seminorm sees windows of fixed length. We prove, unconditionally, that ζ is not Stepanov almost periodic, for any exponent $p \geq 1$, on any closed sub-strip $[\sigma_1, \sigma_2] \subset (\frac{1}{2}, 1)$, in the band (vector-valued) sense: the vectorial Stepanov norm $\sup_x \big(\int_x^{x+1} \int_{\sigma_1}^{\sigma_2} |\zeta(\sigma + it)|^p \, d\sigma \, dt\big)^{1/p}$ is infinite, so $t \mapsto \zeta(\cdot + it)$, viewed as an $L^p(\sigma_1, \sigma_2)$-valued function, is not S^p-almost periodic. The proof combines the unconditional Ω-theorems for large values of ζ on fixed interior lines (Titchmarsh; Aistleitner) with the subharmonicity of $|\zeta|^p$, which promotes a large point value to a large area integral over a band-times-unit-window rectangle. The fixed-line scalar version of the statement is not claimed and remains open; we explain precisely why the method does not decide it. The result places ζ inside a clean dichotomy: the seminorms under which ζ *is* almost periodic in the strip (B^2, Weyl) are blind to compact windows, while the seminorms that see compact windows (uniform, Stepanov) are destroyed by the growth of ζ.

CONTENTS

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1. Introduction

1.1. **Almost periodicity of ζ: the two classical scales.** The connection between the Riemann zeta function and almost periodicity is as old as the theory of almost periodic functions itself: Bohr created the theory in large part as a tool for the Riemann Hypothesis [4]. Two facts anchor the subject.

First, in any closed half-plane $\sigma \geq 1+\delta$ ($\delta > 0$) the Dirichlet series $\sum n^{-s}$ converges absolutely and uniformly, so $t \mapsto \zeta(\sigma+it)$ is a uniform limit of trigonometric polynomials $\sum_{n\leq N} n^{-\sigma} e^{-it\log n}$ and is therefore Bohr (uniformly) almost periodic, on each line and indeed uniformly in the half-plane [4, 3]. This fails on every line $\sigma \leq 1$, simply because $t \mapsto \zeta(\sigma+it)$ is unbounded there (see the Ω-theorems quoted in §3), and Bohr almost periodic functions are bounded.

Second, inside the critical strip the almost periodicity survives in an averaged sense. For fixed $\sigma > \frac{1}{2}$ the partial sums of the Dirichlet series still approximate ζ in quadratic mean over long intervals: by the classical mean-value theorems (Hardy–Littlewood; see [7, Ch. VII]),

$$\limsup_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \Big|\zeta(\sigma+it) - \sum_{n\leq N} n^{-\sigma-it}\Big|^2 dt \xrightarrow[N\to\infty]{} 0,$$

so $t \mapsto \zeta(\sigma+it)$ is almost periodic in the Besicovitch B^2 sense [3, Ch. II] on every line $\sigma > \frac{1}{2}$. Thus ζ is almost periodic in the strip—but only for a seminorm that averages over windows of *growing* length.

1.2. **The intermediate scale, and why it matters.** Between the uniform norm and the Besicovitch seminorm sits the classical hierarchy of almost periodicity classes [3, Ch. II], [5]:

$$\text{Bohr} \ \subset\ S^p \ \subset\ W^p \ \subset\ B^p,$$

where S^p is the Stepanov class, with seminorm $\|f\|_{S^p} = \sup_x \big(\int_x^{x+1} |f|^p\big)^{1/p}$, and W^p, B^p are the Weyl and Besicovitch classes, whose seminorms average over windows of length $L \to \infty$. The Stepanov seminorm is the only intermediate one that *sees windows of fixed length*: it controls the L^p mass of f on every interval of length 1, uniformly in the position of the interval. The Weyl and Besicovitch seminorms do not: for any $f \in L^p_{\text{loc}}$ and any compact K, the Weyl and Besicovitch seminorms of $f \cdot \mathbf{1}_K$ vanish, because the average of a fixed compact contribution over windows of length $L \to \infty$ decays like $1/L$, uniformly in the window position.

This makes the Stepanov scale the natural question for ζ in the strip. Any almost-periodicity statement strong enough to transport local information—values, or zeros via Rouché-type arguments—from one window of the critical strip to another must control fixed windows, which is exactly what B^2 almost periodicity does not do and Stepanov almost periodicity would. The question is therefore: *does the almost periodicity of ζ in the strip survive at the Stepanov scale?*

1.3. **The result.** The answer is no, unconditionally, in the band (vector-valued) sense, and the reason is growth: the same unboundedness that kills Bohr almost periodicity on the lines $\sigma \leq 1$ kills the Stepanov seminorm in the strip, once the pointwise large values are converted into area integrals by subharmonicity.

Theorem (Main theorem; see Theorem 3.3)**.** *Let $B = [\sigma_1, \sigma_2] \subset (\frac{1}{2}, 1)$ with $\sigma_1 < \sigma_2$, and let $p \geq 1$. Then the vectorial Stepanov norm*

$$\|\zeta\|_{S^p(B)} := \sup_{x\in\mathbb{R}} \Big(\int_x^{x+1}\int_{\sigma_1}^{\sigma_2} |\zeta(\sigma+it)|^p \, d\sigma\, dt\Big)^{1/p}$$

is infinite. Consequently $t \mapsto \zeta(\cdot + it)$, viewed as an $L^p(\sigma_1, \sigma_2)$-valued function of $t \in \mathbb{R}$, is not S^p-almost periodic, for any sub-strip B of the open critical strip and any $p \geq 1$.

The statement and proof concern the *band* norm: ζ is treated as a function of the single real variable t with values in the Banach space $L^p(\sigma_1, \sigma_2)$, and the Stepanov seminorm is the classical scalar one with $|f(t)|$ replaced by $\|\zeta(\cdot + it)\|_{L^p(\sigma_1,\sigma_2)}$. This is the standard vector-valued (Banach-space-valued) almost periodicity of Levitan–Zhikov [5]. We emphasise, and discuss in §4, that the fixed-line scalar version—whether $t \mapsto \zeta(\sigma_0 + it)$ can be S^p-almost periodic for some single $\sigma_0 \in (\frac{1}{2}, 1)$—is *not* decided by our method and remains open.

The proof is short and entirely classical in its ingredients: an unconditional Ω-theorem on a fixed interior line (Titchmarsh [7, Ch. VIII]; quantitatively Aistleitner [1]), the sub-mean-value inequality for the subharmonic function $|\zeta|^p$ on disks of a fixed radius contained in the band, and a monotone inclusion of domains. What may be of interest beyond the statement itself is the position the result occupies: together with the classical positive results it exhibits an exact dichotomy in seminorm coordinates (Observation 4.1)—every almost-periodicity seminorm that sees compact windows is destroyed by the growth of ζ in the strip, and every seminorm under which ζ is almost periodic there is blind to compact windows.

2. Stepanov almost periodicity in the band sense

Throughout, $B = [\sigma_1, \sigma_2] \subset (\frac{1}{2}, 1)$ is a closed band with $\sigma_1 < \sigma_2$, and $p \geq 1$.

Definition 2.1 (Vectorial Stepanov norm)**.** For a function $F : \mathbb{R} \to L^p(\sigma_1, \sigma_2)$, locally p-integrable, set

$$\|F\|^p_{S^p(B)} := \sup_{x \in \mathbb{R}} \int_x^{x+1} \|F(t)\|^p_{L^p(\sigma_1,\sigma_2)}\, dt.$$

For $F(t) = \zeta(\cdot + it)$ this is

$$\|\zeta\|^p_{S^p(B)} = \sup_{x \in \mathbb{R}} \int_x^{x+1} \int_{\sigma_1}^{\sigma_2} |\zeta(\sigma + it)|^p \, d\sigma\, dt.$$

Since ζ is holomorphic on the closed band $B \times \mathbb{R}$ (the pole $s = 1$ lies outside B), the map $t \mapsto \zeta(\cdot + it)$ is continuous from $\mathbb{R}$ into $L^p(\sigma_1, \sigma_2)$ (ζ is uniformly continuous on $B \times K$ for every compact $K \subset \mathbb{R}$), so all the integrals above are finite for each fixed x and the supremum is well defined in $[0, \infty]$.

Definition 2.2 (S^p-almost periodicity, band sense)**.** A locally p-integrable $F : \mathbb{R} \to L^p(\sigma_1, \sigma_2)$ is *S^p-almost periodic* if either of the following equivalent conditions holds:

- (i) (closure of trigonometric polynomials) for every $\varepsilon > 0$ there is a trigonometric polynomial $P(t) = \sum_{k=1}^N a_k e^{i\lambda_k t}$ with coefficients $a_k \in L^p(\sigma_1, \sigma_2)$ and real frequencies λ_k such that $\|F - P\|_{S^p(B)} < \varepsilon$;
- (ii) (relatively dense almost periods) for every $\varepsilon > 0$ the set of ε-almost periods $\{\tau \in \mathbb{R} : \|F(\cdot + \tau) - F(\cdot)\|_{S^p(B)} < \varepsilon\}$ is relatively dense in $\mathbb{R}$, i.e. meets every interval of some fixed length $L(\varepsilon)$.

The equivalence of (i) and (ii) is the classical approximation theorem of the Stepanov theory, due in the scalar case to Besicovitch and others [3, Ch. II] and valid verbatim for functions with values in a Banach space [5]; equivalently, the Stepanov theory is the Bohr theory applied to the Bochner transform $t \mapsto F(t + \cdot) \in L^p\big((0, 1); L^p(\sigma_1, \sigma_2)\big)$. We do not reprove the equivalence; the only consequence of almost periodicity we use is the following finiteness lemma, which we prove under *both* definitions so that the main theorem is independent of which one the reader prefers.

Lemma 2.3 (Almost periodicity implies a finite Stepanov norm)**.** *If F is S^p-almost periodic in the sense of Definition 2.2, and $t \mapsto \|F(t)\|_{L^p}$ is locally bounded, then $\|F\|_{S^p(B)} < \infty$.*

Proof. Via (i). Choose P with $\|F-P\|_{S^p(B)} < 1$. A trigonometric polynomial is a bounded function of t with values in $L^p(\sigma_1,\sigma_2)$: $\|P(t)\|_{L^p} \le \sum_k \|a_k\|_{L^p} =: M$ for all t, whence $\|P\|_{S^p(B)} \le M < \infty$. By the triangle inequality for the seminorm $\|\cdot\|_{S^p(B)}$,

$$\|F\|_{S^p(B)} \;\le\; \|F-P\|_{S^p(B)} + \|P\|_{S^p(B)} \;<\; 1+M \;<\; \infty.$$

Via (ii). Take $\varepsilon = 1$ and let $L = L(1)$ be such that every interval of length L contains a 1-almost period. Given $x \in \mathbb{R}$, pick a 1-almost period $\tau \in [x, x+L]$ and set $y := x - \tau \in [-L, 0]$. Then

$$\Big(\int_x^{x+1} \|F(t)\|_{L^p}^p\,dt\Big)^{1/p} = \Big(\int_y^{y+1} \|F(t+\tau)\|_{L^p}^p\,dt\Big)^{1/p} \le 1 + \Big(\int_y^{y+1} \|F(t)\|_{L^p}^p\,dt\Big)^{1/p},$$

and the last quantity is bounded by $1 + \sup_{y\in[-L,0]}\big(\int_y^{y+1}\|F\|_{L^p}^p\big)^{1/p}$, which is finite by local boundedness. The bound is uniform in x. □

Note that the conclusion is finiteness of the *norm* $\|F\|_{S^p(B)}$, not pointwise boundedness of F: an S^p limit of bounded functions need not be pointwise bounded, but its S^p norm is finite, and that is all the main theorem contradicts. For $F(t) = \zeta(\cdot + it)$ the local boundedness hypothesis holds, since ζ is continuous on the closed band.

3. The main theorem

The analytic input is an unconditional lower bound for large values of ζ on a fixed line in the interior of the strip. Two classical sources suffice, and either one alone is enough for our purposes.

Theorem 3.1 (Titchmarsh [7, Ch. VIII])**.** *For fixed $\sigma \in (\frac12, 1)$ and every $\varepsilon > 0$,*

$$|\zeta(\sigma+it)| = \Omega\Big(\exp\big((\log t)^{1-\sigma-\varepsilon}\big)\Big) \qquad (t\to\infty),$$

unconditionally. In particular $t \mapsto \zeta(\sigma+it)$ is unbounded on every such line.

Theorem 3.2 (Aistleitner [1])**.** *For fixed $\alpha \in (\frac12, 1)$ there is a constant $c_\alpha = 0.18\,(2\alpha-1)^{1-\alpha}$ such that, unconditionally, for all sufficiently large T,*

$$\max_{0\le t\le T} |\zeta(\alpha+it)| \;\ge\; \exp\Big(c_\alpha\,\frac{(\log T)^{1-\alpha}}{(\log\log T)^{\alpha}}\Big).$$

Theorem 3.2 sharpens, by the resonance method, a result going back to Montgomery [6]; we stress that the unconditional statements we rely on are Theorems 3.1 and 3.2 as stated, and that the quantitative rates play no role below: the proof needs only the qualitative conclusion that ζ is unbounded on one interior line of each sub-band.

Theorem 3.3 (Main theorem)**.** *Let $B = [\sigma_1,\sigma_2] \subset (\frac12, 1)$ with $\sigma_1 < \sigma_2$, and let $p \ge 1$. Then*

$$\|\zeta\|_{S^p(B)} = \infty.$$

In particular, by Lemma 2.3, $t \mapsto \zeta(\cdot + it)$ is not S^p-almost periodic as an $L^p(\sigma_1,\sigma_2)$-valued function, under either definition in Definition 2.2.

Proof. Fix $\sigma \in (\sigma_1, \sigma_2)$ and set

$$r := \tfrac12 \min\big(\sigma-\sigma_1,\ \sigma_2-\sigma,\ \tfrac14\big) > 0,$$

so that $r \le \frac18$ and every disk $D(\sigma+it, r)$ with $t \in \mathbb{R}$ satisfies the inclusion

$$D(\sigma+it,\, r) \;\subset\; [\sigma_1,\sigma_2]\times[t-r,\, t+r], \tag{1}$$

because $r \le (\sigma-\sigma_1)/2$ and $r \le (\sigma_2-\sigma)/2$.

Step 1: a sequence of large values. By Theorem 3.1 (or Theorem 3.2) there exists a sequence $t_n \to \infty$ with

$$V_n := |\zeta(\sigma+it_n)| \longrightarrow \infty.$$

For an Ω-statement this is immediate (a sequence realising the lim sup). For a statement of the form $\max_{0\le t\le T}|\zeta(\sigma+it)| \ge \Phi(T) \to \infty$, let t_T be a maximiser on $[0,T]$; the t_T cannot remain in a compact set, since ζ is continuous, hence bounded, on $[\sigma_1,\sigma_2]\times[0,K]$ for every K, while $|\zeta(\sigma+it_T)| \ge \Phi(T)\to\infty$; so $t_T\to\infty$ along a subsequence, which we take as (t_n). We may also assume $t_n>1$ for all n, so that the disks below stay away from the real axis.

Step 2: subharmonicity. The function $|\zeta|^p$ is subharmonic on a neighbourhood of $B\times[1,\infty)$: ζ is holomorphic there, $\log|\zeta|$ is subharmonic, and $|\zeta|^p = \exp(p\log|\zeta|)$ is the composition of a subharmonic function with the convex increasing function $\exp(p\,\cdot)$, hence subharmonic (this holds for every $p>0$; the restriction $p\ge 1$ only fixes the classical Stepanov class). The sub-mean-value inequality on the disk $D(\sigma+it_n,r)$ gives

$$V_n^p \;\le\; \frac{1}{\pi r^2}\iint_{D(\sigma+it_n,\,r)} |\zeta|^p\,dA.$$

Step 3: inclusion of domains. The integrand is nonnegative, so by (1),

$$V_n^p \;\le\; \frac{1}{\pi r^2}\int_{t_n-r}^{t_n+r}\int_{\sigma_1}^{\sigma_2} |\zeta(\sigma'+it)|^p\,d\sigma'\,dt.$$

Step 4: the window fits. Since $2r\le\frac14<1$, the time interval $[t_n-r,t_n+r]$ is contained in the unit window $[x_n,x_n+1]$ with $x_n:=t_n-r$. Hence

$$\|\zeta\|_{S^p(B)}^p \;\ge\; \int_{x_n}^{x_n+1}\int_{\sigma_1}^{\sigma_2} |\zeta(\sigma'+it)|^p\,d\sigma'\,dt \;\ge\; \pi r^2\,V_n^p \;\longrightarrow\; \infty,$$

and the supremum over x is infinite. Each individual window integral is finite (ζ is continuous on $B\times$ compacts); the divergence is genuinely a divergence of the supremum along $x_n\to\infty$, as it must be. ☐

Remark 3.4 (What the proof does and does not use). (a) The argument never converts the two-dimensional sub-mean value into a one-dimensional average at a fixed abscissa: the quantity shown to be infinite *is* the two-dimensional band-times-window average, and Step 3 is a monotone inclusion of domains with a nonnegative integrand—no Fubini, no choice of a distinguished vertical line inside the disk. This is precisely why the theorem is a statement about the band norm and not about any fixed line; see Problem 4.2. (b) The growth *rate* in Theorems 3.1–3.2 is irrelevant: S^p-almost periodicity would impose one fixed finite bound $\|\zeta\|_{S^p(B)}=C<\infty$, and any sequence $V_n\to\infty$ violates $\pi r^2 V_n^p\le C^p$. Mere unboundedness of ζ on a single interior line of the sub-band suffices. (c) The proof works for every $p>0$ and every sub-band, with the radius r calibrated locally; the hypothesis $\sigma_1<\sigma_2$ is essential (for $\sigma_1=\sigma_2$ there is no band and no disk).

Remark 3.5 (Non-membership, not just non-almost-periodicity). The theorem proves more than the failure of almost periodicity: $\|\zeta\|_{S^p(B)}=\infty$ means ζ does not even belong to the Stepanov space on which the class of S^p-almost periodic functions is modelled. The prose statement "ζ is not S^p-almost periodic on any sub-strip, in the band sense" is the correct consequence via Lemma 2.3.

4. Discussion

4.1. **A dichotomy in seminorm coordinates.**

Observation 4.1 (Seeing windows versus surviving growth). In the strip $\frac12<\sigma<1$, the classical almost-periodicity seminorms split into two groups, and ζ separates them exactly.

(a) *Seminorms blind to compact windows (B^p, W^p):* the Besicovitch and Weyl seminorms average over windows of length $L\to\infty$, so the seminorm of $f\cdot\mathbf{1}_K$ vanishes for every compact K (for Weyl, note that the supremum over the window position is taken *before* the limit $L\to\infty$, and the average of a fixed compact contribution still decays like $1/L$ uniformly in the position). Under these seminorms ζ *is* almost periodic in the strip (B^2, classically, from the mean-square approximation by partial sums; §1).

(b) *Seminorms that see compact windows (uniform, S^p):* the uniform norm and the Stepanov seminorms control fixed windows uniformly in their position. Under these, almost periodicity of ζ fails in the strip: uniformly because ζ is unbounded on every line $\sigma < 1$ (Theorem 3.1), and in the Stepanov band sense by Theorem 3.3.

In short: *every seminorm under which ζ is almost periodic in the strip is blind to compact windows, and every seminorm that sees compact windows is destroyed by the growth of ζ.* The mechanism of failure is in both cases growth, not arithmetic: the unconditional large values of ζ are incompatible with any almost-periodicity seminorm that assigns mass to windows of fixed length.

4.2. **The open fixed-line problem.**

Problem 4.2 (Fixed-line scalar Stepanov almost periodicity)**.** Is $t \mapsto \zeta(\sigma_0+it)$ S^p-almost periodic, as a scalar function, for some $\sigma_0 \in (\frac{1}{2}, 1)$ and some $p \geq 1$? Equivalently, in the bounded direction: is $\sup_x \int_x^{x+1} |\zeta(\sigma_0 + it)|^p \, dt$ finite for some such σ_0?

We claim nothing about this problem; in particular, Theorem 3.3 does *not* decide it, and the reason is structural. The sub-mean-value inequality spreads the mass of a large point value over a two-dimensional neighbourhood, not over the line through the point: from $|\zeta(\sigma_0 + it_n)|$ large one can only conclude that the area integral over a disk is large, hence that *some* abscissa σ_n'' (depending on n) carries a large one-dimensional window integral—and there is no way, within this argument, to pin the moving σ_n'' to the fixed line σ_0. Nor do the classical convexity tools apply: the logarithmic convexity in σ of L^p means of holomorphic functions on vertical lines is a statement about integrals over the whole line, not over unit windows, and there is no one-dimensional sub-mean-value inequality for $|g|^p$, g holomorphic, along a vertical segment. The band formulation of Theorem 3.3 is not a technical convenience but the exact boundary of what the subharmonicity method proves.

4.3. **Relation to the Riemann Hypothesis.** The result is a negative theorem about seminorms and is logically independent of the Riemann Hypothesis (it is unconditional). Its contextual relevance is this: by a classical theme going back to Bohr [4] and made precise by Bagchi [2], RH is equivalent to a strong recurrence (self-approximation) property of ζ in compact subsets of the strip, so one might hope to approach such recurrence statements through an almost-periodicity seminorm strong enough to control compact windows. Theorem 3.3 shows that the Stepanov scale—the only classical intermediate scale with that property—is unconditionally unavailable in the band sense; any window-localised recurrence for ζ must therefore be obtained by mechanisms other than membership of ζ in a classical almost-periodicity class that sees windows. We make no claim beyond this.

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